

Hard-collision robustness metrics

Continuous values: mean +/- sample SD; control roughness is mean per transition (lower is smoother); roughness/effort/steps: successful trials only

Case / variant	trials	Success	Control roughness overall (mean/transition)	Control roughness after wall (mean/transition)	Control effort	Number of steps	Average time per step (s)
condition=dynamic_wall variant=control_bank_mpp	40	30/40 (75.0%)	0.635 +/- 0.211 (n=30)	0.716 +/- 0.216 (n=30)	180.355 +/- 22.569 (n=30)	75.367 +/- 12.053 (n=30)	0.048 +/- 0.012 (n=40)
condition=dynamic_wall variant=corridor_prior_map	40	28/40 (70.0%)	1.006 +/- 0.368 (n=28)	1.106 +/- 0.373 (n=28)	183.576 +/- 21.209 (n=28)	89.250 +/- 13.729 (n=28)	0.042 +/- 0.009 (n=40)
condition=dynamic_wall variant=empirical_init_map	40	30/40 (75.0%)	0.635 +/- 0.211 (n=30)	0.716 +/- 0.216 (n=30)	180.355 +/- 22.569 (n=30)	75.367 +/- 12.053 (n=30)	0.045 +/- 0.011 (n=40)
condition=dynamic_wall variant=frenet_corridor_map	40	38/40 (95.0%)	0.998 +/- 0.417 (n=38)	1.148 +/- 0.516 (n=38)	186.999 +/- 18.648 (n=38)	87.447 +/- 9.888 (n=38)	0.042 +/- 0.009 (n=40)
condition=dynamic_wall variant=full_swarm_prior_map	40	26/40 (65.0%)	0.785 +/- 0.278 (n=26)	0.847 +/- 0.336 (n=26)	174.361 +/- 23.694 (n=26)	94.962 +/- 12.568 (n=26)	0.040 +/- 0.008 (n=40)
condition=dynamic_wall variant=gaussian_prior_map	40	25/40 (62.5%)	0.522 +/- 0.153 (n=25)	0.555 +/- 0.154 (n=25)	166.715 +/- 22.148 (n=25)	94.000 +/- 9.372 (n=25)	0.007 +/- 0.001 (n=40)
condition=dynamic_wall variant=heatmap_prior_map	40	25/40 (62.5%)	0.690 +/- 0.219 (n=25)	0.691 +/- 0.239 (n=25)	174.890 +/- 24.162 (n=25)	90.000 +/- 16.894 (n=25)	0.042 +/- 0.009 (n=40)
condition=dynamic_wall variant=homotopy_seeded	40	26/40 (65.0%)	0.702 +/- 0.228 (n=26)	0.698 +/- 0.266 (n=26)	183.896 +/- 28.018 (n=26)	82.192 +/- 19.539 (n=26)	0.042 +/- 0.009 (n=40)
condition=dynamic_wall variant=mode_selecting_corridor	40	25/40 (62.5%)	0.858 +/- 0.244 (n=25)	0.908 +/- 0.254 (n=25)	180.034 +/- 20.401 (n=25)	87.200 +/- 9.789 (n=25)	0.024 +/- 0.002 (n=40)
condition=dynamic_wall variant=mode_selecting_homotopy	40	25/40 (62.5%)	0.696 +/- 0.290 (n=25)	0.773 +/- 0.364 (n=25)	180.355 +/- 24.948 (n=25)	77.040 +/- 10.342 (n=25)	0.023 +/- 0.002 (n=40)
condition=dynamic_wall variant=standard_mpp	40	21/40 (52.5%)	0.209 +/- 0.094 (n=21)	0.176 +/- 0.056 (n=21)	153.390 +/- 9.304 (n=21)	70.048 +/- 13.927 (n=21)	0.003 +/- 0.000 (n=40)
condition=no_wall variant=control_bank_map	10	10/10 (100.0%)	0.591 +/- 0.158 (n=10)	-	166.645 +/- 9.523 (n=10)	68.800 +/- 2.440 (n=10)	0.048 +/- 0.012 (n=10)
condition=no_wall variant=corridor_prior_map	10	10/10 (100.0%)	0.972 +/- 0.445 (n=10)	-	170.467 +/- 10.726 (n=10)	81.700 +/- 4.692 (n=10)	0.043 +/- 0.009 (n=10)
condition=no_wall variant=empirical_init_map	10	10/10 (100.0%)	0.591 +/- 0.158 (n=10)	-	166.645 +/- 9.523 (n=10)	68.800 +/- 2.440 (n=10)	0.046 +/- 0.012 (n=10)
condition=no_wall variant=frenet_corridor_map	10	10/10 (100.0%)	0.872 +/- 0.440 (n=10)	-	180.186 +/- 11.803 (n=10)	83.300 +/- 6.038 (n=10)	0.043 +/- 0.009 (n=10)
condition=no_wall variant=full_swarm_prior_map	10	10/10 (100.0%)	0.695 +/- 0.247 (n=10)	-	158.904 +/- 7.839 (n=10)	90.200 +/- 5.029 (n=10)	0.042 +/- 0.011 (n=10)
condition=no_wall variant=gaussian_prior_map	10	10/10 (100.0%)	0.487 +/- 0.158 (n=10)	-	154.689 +/- 11.637 (n=10)	90.100 +/- 4.557 (n=10)	0.007 +/- 0.001 (n=10)
condition=no_wall variant=heatmap_prior_map	10	9/10 (90.0%)	0.597 +/- 0.145 (n=9)	-	156.534 +/- 7.147 (n=9)	79.556 +/- 4.475 (n=9)	0.042 +/- 0.009 (n=10)
condition=no_wall variant=homotopy_seeded	10	9/10 (90.0%)	0.660 +/- 0.161 (n=9)	-	165.273 +/- 5.504 (n=9)	70.000 +/- 2.236 (n=9)	0.043 +/- 0.010 (n=10)
condition=no_wall variant=mode_selecting_corridor	10	10/10 (100.0%)	0.839 +/- 0.269 (n=10)	-	171.795 +/- 8.637 (n=10)	83.100 +/- 3.479 (n=10)	0.024 +/- 0.002 (n=10)
condition=no_wall variant=mode_selecting_homotopy	10	10/10 (100.0%)	0.592 +/- 0.230 (n=10)	-	168.282 +/- 16.759 (n=10)	71.700 +/- 5.334 (n=10)	0.023 +/- 0.002 (n=10)
condition=no_wall variant=standard_mpp	10	10/10 (100.0%)	0.197 +/- 0.079 (n=10)	-	150.584 +/- 6.411 (n=10)	67.400 +/- 11.217 (n=10)	0.003 +/- 0.001 (n=10)
condition=static_wall variant=control_bank_map	40	38/40 (95.0%)	0.602 +/- 0.245 (n=38)	0.602 +/- 0.245 (n=38)	181.090 +/- 31.837 (n=38)	74.079 +/- 9.598 (n=38)	0.027 +/- 0.007 (n=40)
condition=static_wall variant=corridor_prior_map	40	40/40 (100.0%)	0.914 +/- 0.317 (n=40)	0.914 +/- 0.317 (n=40)	178.834 +/- 15.210 (n=40)	86.925 +/- 5.740 (n=40)	0.024 +/- 0.006 (n=40)
condition=static_wall variant=empirical_init_map	40	38/40 (95.0%)	0.602 +/- 0.245 (n=38)	0.602 +/- 0.245 (n=38)	181.090 +/- 31.837 (n=38)	74.079 +/- 9.598 (n=38)	0.025 +/- 0.007 (n=40)
condition=static_wall variant=frenet_corridor_map	40	40/40 (100.0%)	0.927 +/- 0.323 (n=40)	0.927 +/- 0.323 (n=40)	187.215 +/- 15.212 (n=40)	85.575 +/- 5.905 (n=40)	0.024 +/- 0.006 (n=40)
condition=static_wall variant=full_swarm_prior_map	40	40/40 (100.0%)	0.715 +/- 0.313 (n=40)	0.715 +/- 0.313 (n=40)	170.534 +/- 14.212 (n=40)	96.750 +/- 5.861 (n=40)	0.023 +/- 0.006 (n=40)
condition=static_wall variant=gaussian_prior_map	40	40/40 (100.0%)	0.526 +/- 0.154 (n=40)	0.526 +/- 0.154 (n=40)	168.503 +/- 23.439 (n=40)	96.975 +/- 9.420 (n=40)	0.005 +/- 0.001 (n=40)
condition=static_wall variant=heatmap_prior_map	40	39/40 (97.5%)	0.661 +/- 0.275 (n=39)	0.661 +/- 0.275 (n=39)	169.537 +/- 18.282 (n=39)	84.359 +/- 6.037 (n=39)	0.024 +/- 0.006 (n=40)
condition=static_wall variant=homotopy_seeded	40	40/40 (100.0%)	0.740 +/- 0.353 (n=40)	0.740 +/- 0.353 (n=40)	181.042 +/- 33.986 (n=40)	76.125 +/- 9.851 (n=40)	0.024 +/- 0.006 (n=40)
condition=static_wall variant=mode_selecting_corridor	40	40/40 (100.0%)	0.845 +/- 0.186 (n=40)	0.845 +/- 0.186 (n=40)	178.142 +/- 11.357 (n=40)	85.225 +/- 4.917 (n=40)	0.022 +/- 0.003 (n=40)
condition=static_wall variant=mode_selecting_homotopy	40	40/40 (100.0%)	0.711 +/- 0.306 (n=40)	0.711 +/- 0.306 (n=40)	177.956 +/- 20.038 (n=40)	74.150 +/- 5.929 (n=40)	0.022 +/- 0.002 (n=40)
condition=static_wall variant=standard_mpp	40	25/40 (62.5%)	0.227 +/- 0.102 (n=25)	0.227 +/- 0.102 (n=25)	154.815 +/- 10.770 (n=25)	71.120 +/- 13.803 (n=25)	0.002 +/- 0.000 (n=40)

Failure reasons

Counts and percentages among failed trials

Case / variant	Failure reasons
condition=dynamic_wall variant=control_bank_mppi	not_reaching: 10 (100.0%)
condition=dynamic_wall variant=corridor_prior_mppi	not_reaching: 12 (100.0%)
condition=dynamic_wall variant=empirical_init_mppi	not_reaching: 10 (100.0%)
condition=dynamic_wall variant=frenet_corridor_mppi	not_reaching: 2 (100.0%)
condition=dynamic_wall variant=full_swarm_prior_mppi	not_reaching: 14 (100.0%)
condition=dynamic_wall variant=gaussian_prior_mppi	not_reaching: 15 (100.0%)
condition=dynamic_wall variant=heatmap_prior_mppi	not_reaching: 15 (100.0%)
condition=dynamic_wall variant=homotopy_seeded_mppi	not_reaching: 14 (100.0%)
condition=dynamic_wall variant=mode_selecting_corridor_mppi	not_reaching: 15 (100.0%)
condition=dynamic_wall variant=mode_selecting_homotopy_mppi	not_reaching: 15 (100.0%)
condition=dynamic_wall variant=standard_mppi	not_reaching: 19 (100.0%)
condition=no_wall variant=control_bank_mppi	none
condition=no_wall variant=corridor_prior_mppi	none
condition=no_wall variant=empirical_init_mppi	none
condition=no_wall variant=frenet_corridor_mppi	none
condition=no_wall variant=full_swarm_prior_mppi	none
condition=no_wall variant=gaussian_prior_mppi	none
condition=no_wall variant=heatmap_prior_mppi	not_reaching: 1 (100.0%)
condition=no_wall variant=homotopy_seeded_mppi	not_reaching: 1 (100.0%)
condition=no_wall variant=mode_selecting_corridor_mppi	none
condition=no_wall variant=mode_selecting_homotopy_mppi	none
condition=no_wall variant=standard_mppi	none
condition=static_wall variant=control_bank_mppi	not_reaching: 2 (100.0%)
condition=static_wall variant=corridor_prior_mppi	none
condition=static_wall variant=empirical_init_mppi	not_reaching: 2 (100.0%)
condition=static_wall variant=frenet_corridor_mppi	none
condition=static_wall variant=full_swarm_prior_mppi	none
condition=static_wall variant=gaussian_prior_mppi	none
condition=static_wall variant=heatmap_prior_mppi	not_reaching: 1 (100.0%)
condition=static_wall variant=homotopy_seeded_mppi	none
condition=static_wall variant=mode_selecting_corridor_mppi	none
condition=static_wall variant=mode_selecting_homotopy_mppi	none
condition=static_wall variant=standard_mppi	not_reaching: 15 (100.0%)